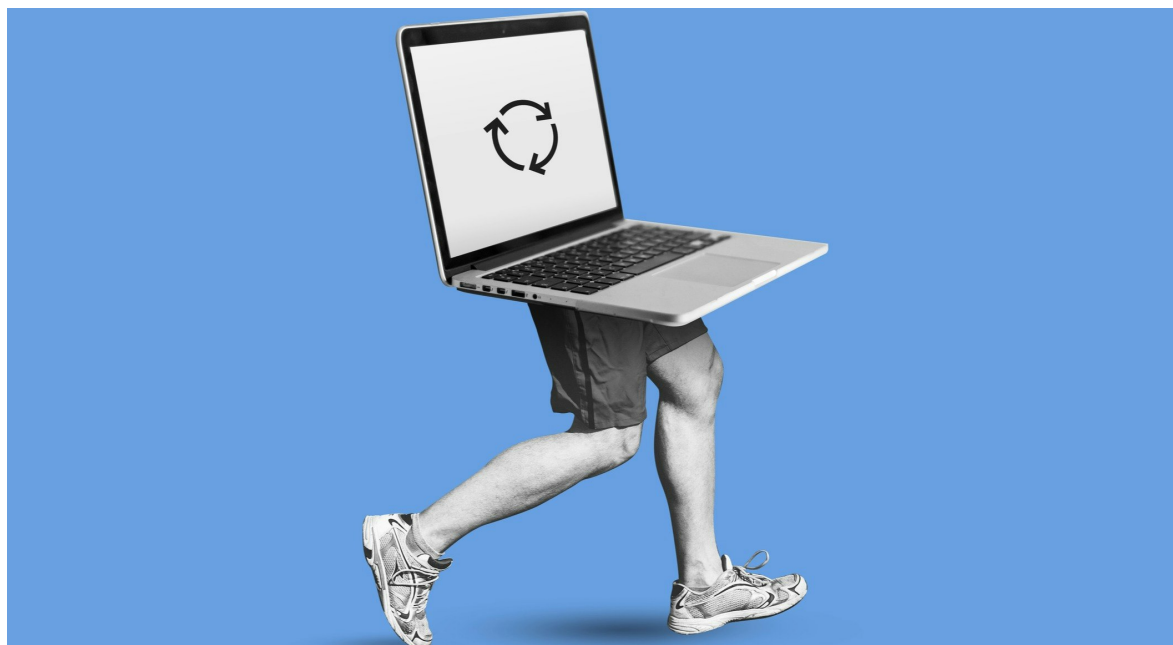




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Performance testing 5m



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Performance testing is vital to measure how well a product performs from an end-to-end perspective.

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It's a type of nonfunctional testing used to:

- demonstrate a system meets predetermined performance criteria
- compare the performance of two systems
- determine which parts of a system are performing badly and why.

Performance testing metrics

Let's look at some commonly tested performance features and the metrics associated with them in more detail:

- **Load size:** the number of concurrent users at any one point in time. This is also called a stress test.
 - **Response time:** the time that passes from when a user sends a request to when the application has completed processing the request. It's measured in seconds but often reported in aggregated numbers such as the average, 90th percentile or peak response time (the longest time to fulfil a request).
 - **Wait time:** the time taken to receive the first byte of information after a request is sent. It's also known as the average latency.
 - **Load time:** the amount of time to deliver every request. It's often presented as an average.
 - **Throughput:**
the number of pages or requests that are hitting the server in one second. It's determined by the amount of traffic on the application at any one point in time and can indicate the amount of bandwidth used during the test. It's measured in hits per second or kilobytes per second.
 - **Transactions passed or failed:** the total number of successful or unsuccessful requests.
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- **Error rate:** a key metric to determine the stability of an application. It's measured in errors per second or as a percentage of the total requests.

On the server side, these are typical metrics measured in performance testing:

- **CPU:** CPU percentage utilisation or how much time the CPU needs to process the requests.
- **Memory:** memory usage during a test.
- **Disk:** disk space utilised during a test.

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Performance testing will also give you information about generic characteristics such as:

- **Stability:** the ability of an application or system to continue to function effectively over time without failing.
- **Scalability:** the ability of an application or system to continue to function effectively as demand or usage increases.
- **Availability:** the amount of time a device is actually operating as a percentage of the time it should be operating without any outages. It's typically measured in percentages and can be translated into hours using <https://uptime.is/>.
- **Utilisation:** the amount of time a component is processing. Generally, the higher the percentage, the better the component is being utilised, but if utilisation is too high (for example, over 90%) performance may suffer, and reasons for excessive utilisation should be investigated.
- **Reliability:** the ability of an application or system to function effectively within a specified time and environment without failing.

Remember: Performance testing cannot simulate every possible combination of business activity. So, the Pareto principle, or 80/20 observation, is often applied. This principle illustrates that 80% of effects, outputs or results are obtained from, or due to, 20% of causes, inputs or effort.

So, in performance testing, a set of frequently performed, significant transactions or activities is selected (these are the 20%) that should provide 80% of the results needed. The effort and focus should be in these areas.



Thoughts on this page?



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